



University of Pennsylvania | 04/22/2025



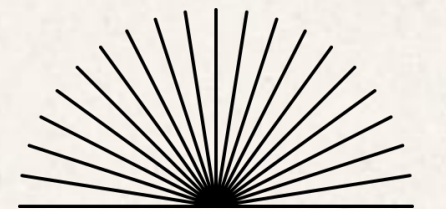
AGRICULTURAL MAPPING AT SCALE

Synthetic Image Generation For LATAM Field Segmentation

NAME OF PROJECT:
Agricultural Mapping at Scale

PRESENTED BY:
Muhammad Abdul Rahman |
Jiayue Ma

PRESENTED TO:
MUSA 6500
Deep Learning in Python

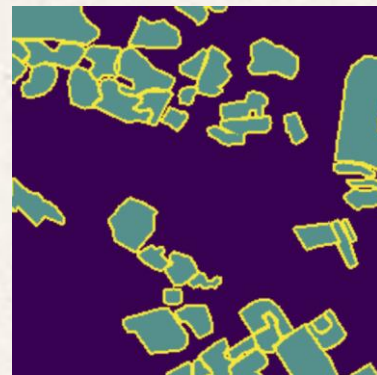


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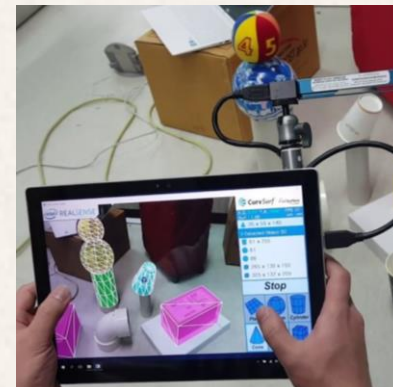
Problem Identification

Current Bottlenecks in Latin American (LATAM) Agricultural Field Segmentation



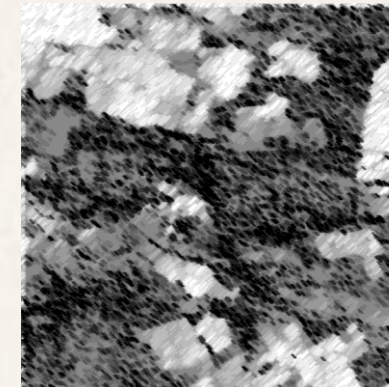
Limited Labeled Data in LATAM

Too few annotated images—lacking regional variability.



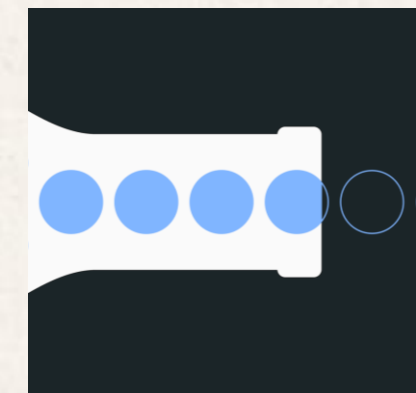
High Cost and Time of Manual Annotation

Pixel-perfect masks are slow and expensive.



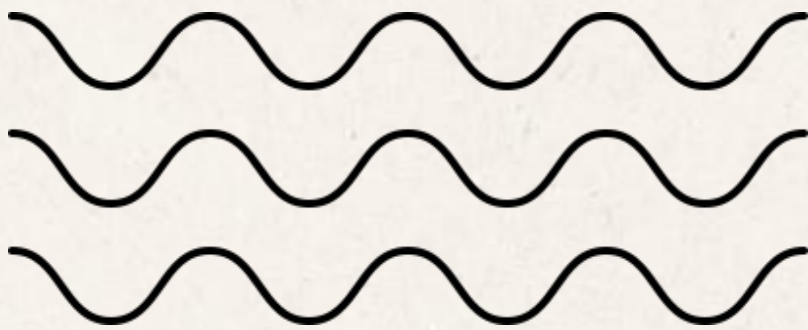
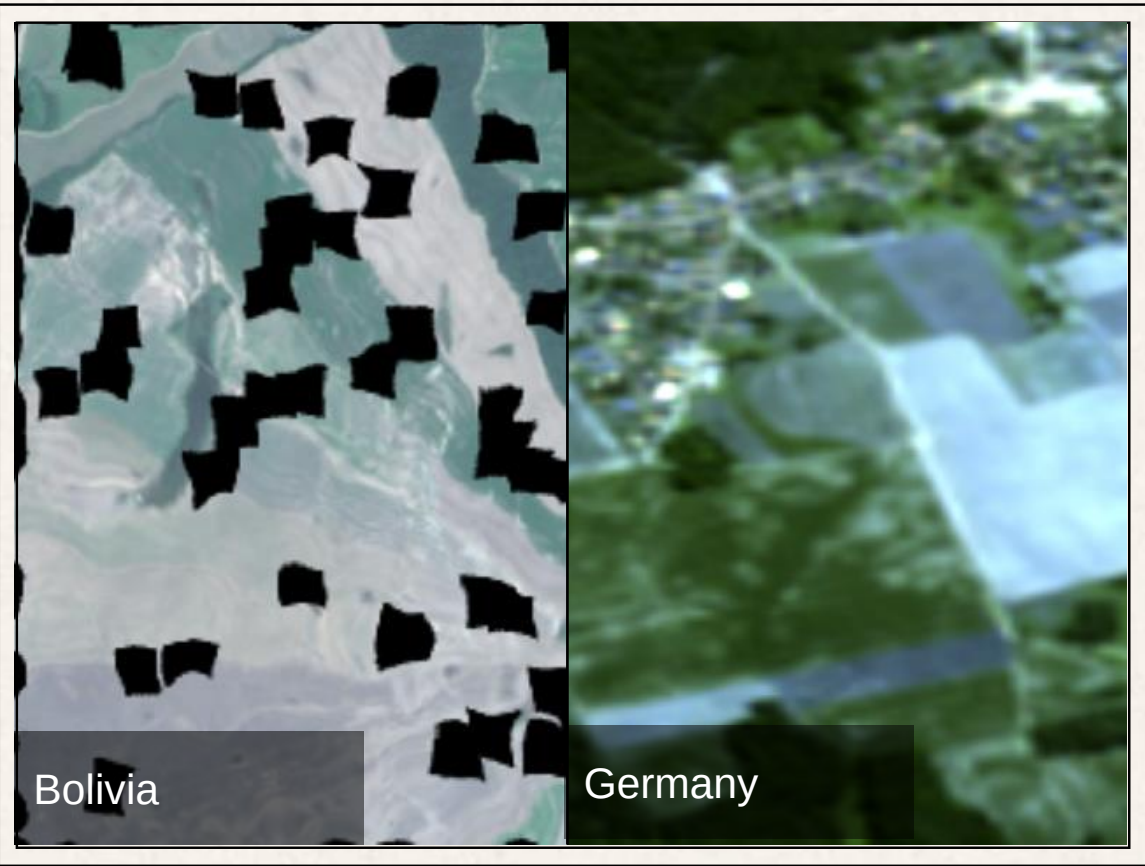
Insufficient Consideration for “Bad” Data Scenarios

Cloud cover, sensor dropouts and missing tiles are under-represented.



Model Performance Bottleneck?

- Low accuracy on unseen LATAM tiles
- Smallholder vs. industrial plot errors
- Seasonal sensitivity (bare soil vs. dense canopy)



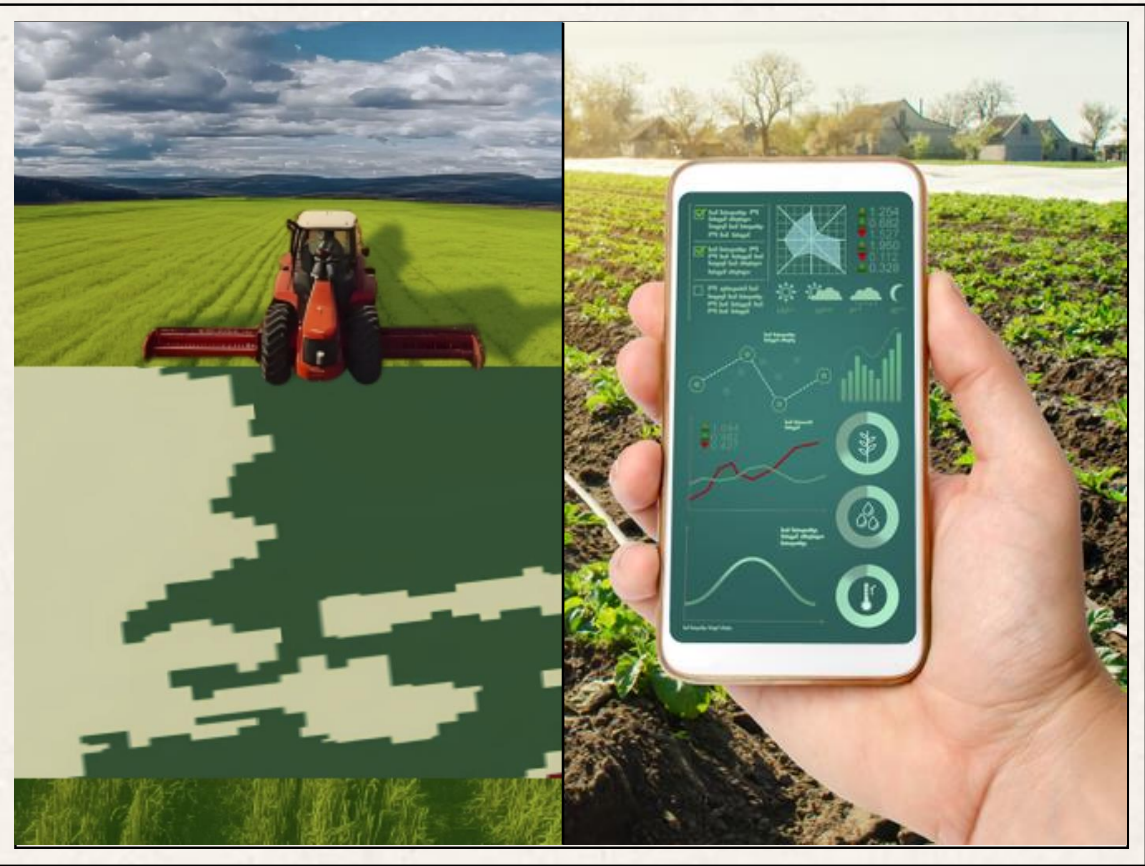
Proposed Solution & Target Audience

Synthetic Imagery for Data Augmentation

To overcome our labeling limits and domain gaps, we'll use two methods:

- **Augmented Original Dataset:** Apply both classical/non-classical (rotations, flips, dropout, elastic deforms) and generative (GMM or GAN) transforms to our LATAM labels—creating a wide array of synthetic samples.
- **Curated External Imagery:** Import satellite data from other regions and adapt their bands, resolution, and annotations to LATAM style using above augmentations —expanding diversity without extra manual labeling.

The potential **Target Audience** for this application include the data scientists fine-tuning field-segmentation models with limited labels, and the agricultural analysts who rely on accurate parcel maps for yield forecasts and management decisions. By boosting data diversity without a huge jump in manual work, we hope to help both sides get more reliable, scalable insights across LATAM's farms.

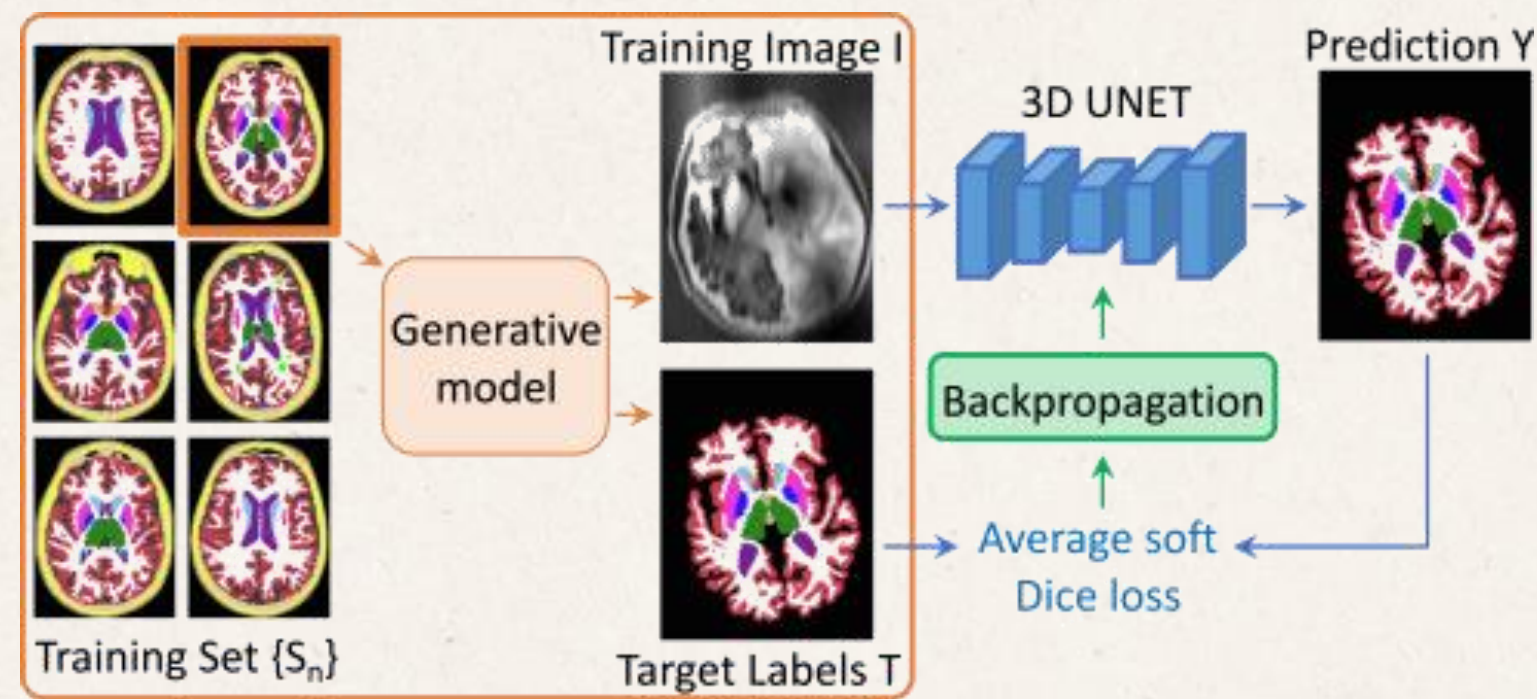


Crucial Steps in Generative Algorithm:

- Random Affine Transformation / Elastic Wrap
- Intensity Sampling (Color Image by Class)
- Bias-Field Perturbation (Generate a Spotlight Pattern)
- Add Noise or Artifacts

Takeaways for Our Proposal:

- Add Random Affine Transformation / Elastic Wrap to Mimic Distortions in Satellite Imagery
- Add Noise to Mimic Cloud Cover/Blur in Satellite Imagery
- Crop out random extents to Mimic Irregular Shape of Fields



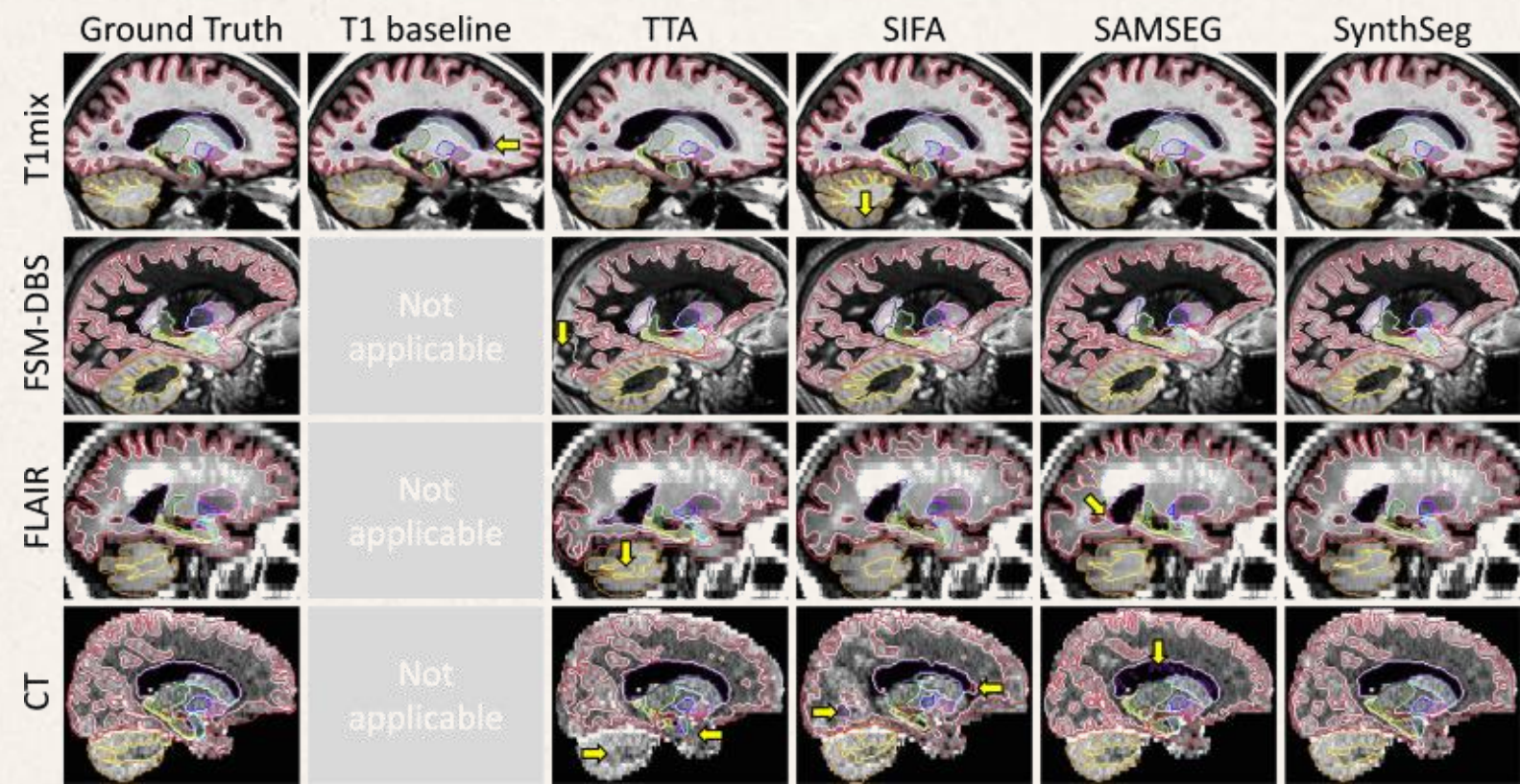
Relevant Research

SynthSeg

(For Brain MRI Images)

SynthSeg trains a convolutional network solely on synthetic MRI scans generated from anatomical label maps. It learns features that generalize across any real scanner or protocol—achieving state-of-the-art segmentation without touching a real image during training.

Benjamin Billot et al.,
SynthSeg: Segmentation of brain MRI scans of any contrast and resolution without retraining, Medical Image Analysis, 2023



Data

Baseline Dataset that We Work With

Data Source

We use a Bolivia dataset sourced from the World Resources Institute (WRI), and an additional Germany dataset downloaded from Fields of the World (FTW) benchmark agricultural dataset. Each dataset contains 85 pairs of images.

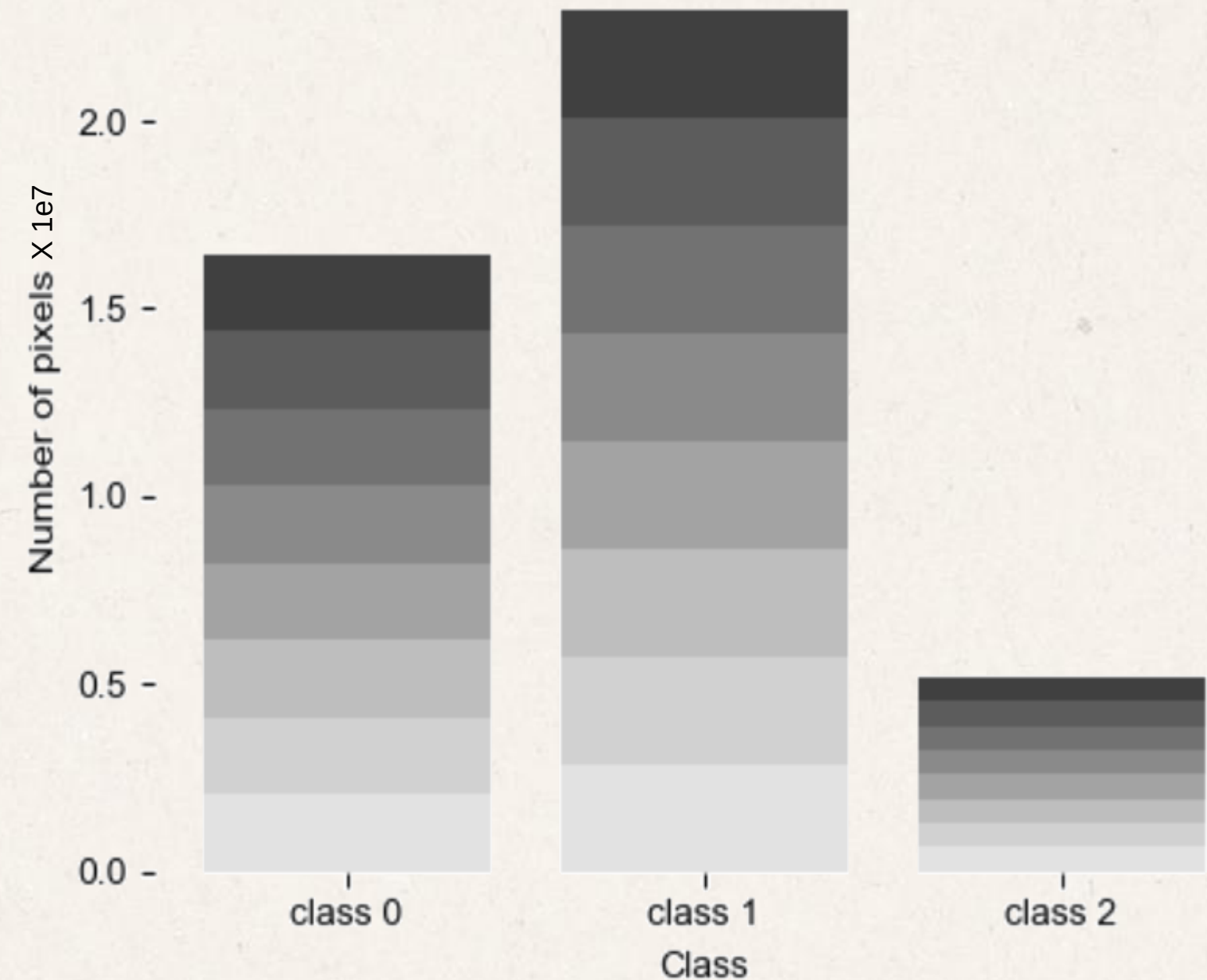
Data Type and Dimension

The dataset contains of paired sentinel-2 satellite images and 3-class label masks (Background (0), Field Interior (1) and Field Boundary (2).). All images are in 256*256 pixels.

Bands Information

The satellite image contains 8 bands: RGB + NIR from window a and RGB + NIR from window b.

Bolivia Dataset Pixel Count By Class

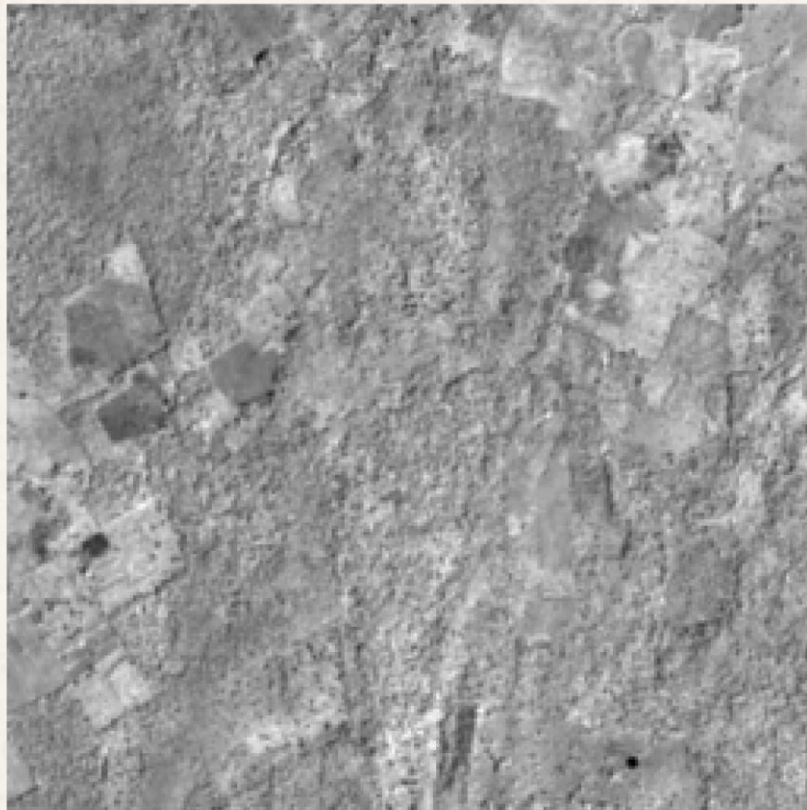


Data

Baseline Dataset that We Work With

Bolivia Dataset

Window A

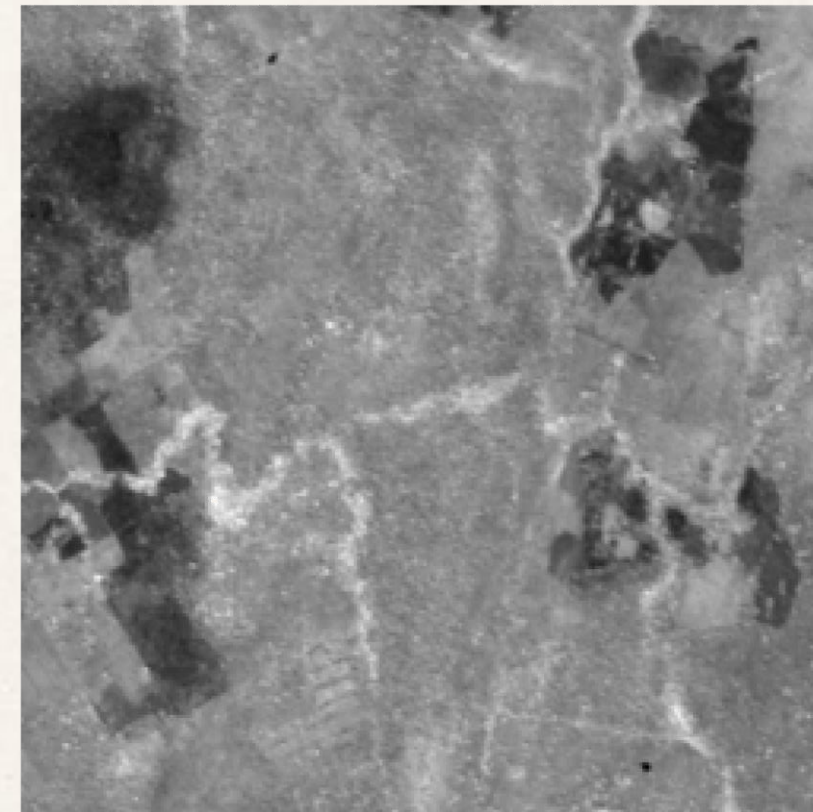


NIR



RGB

Window B



NIR



RGB

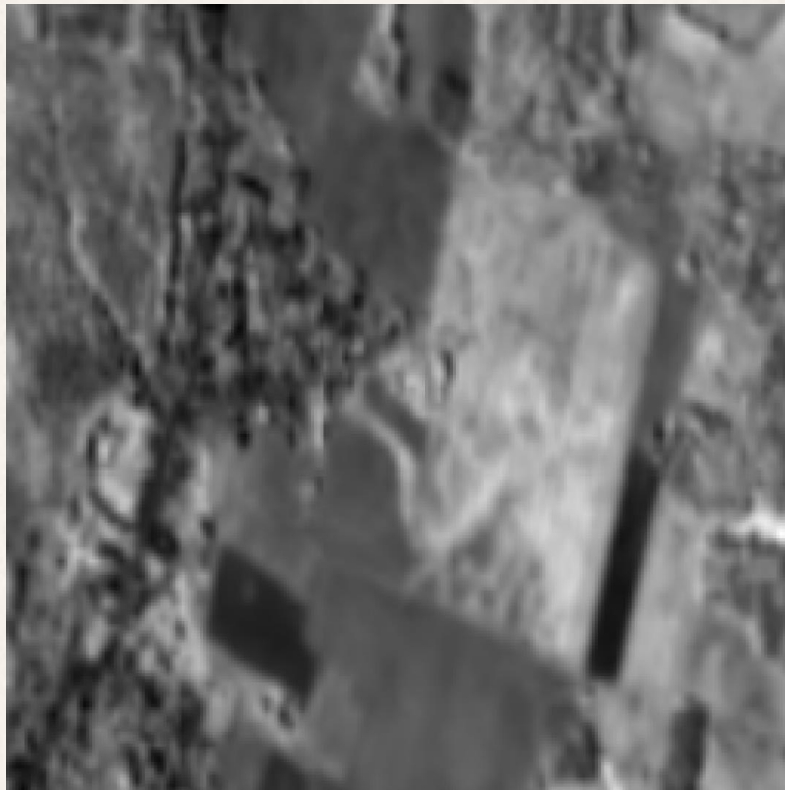
(Window a and b refer to different growing seasons which are combined when fed to the model.)

Data

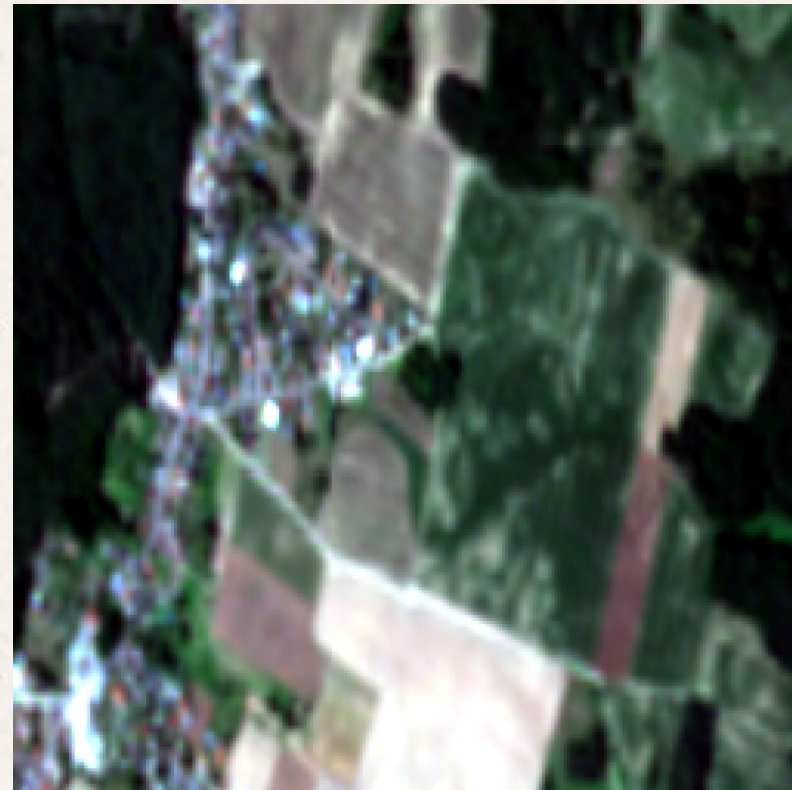
Baseline Dataset that We Work With

Germany Dataset

Window A

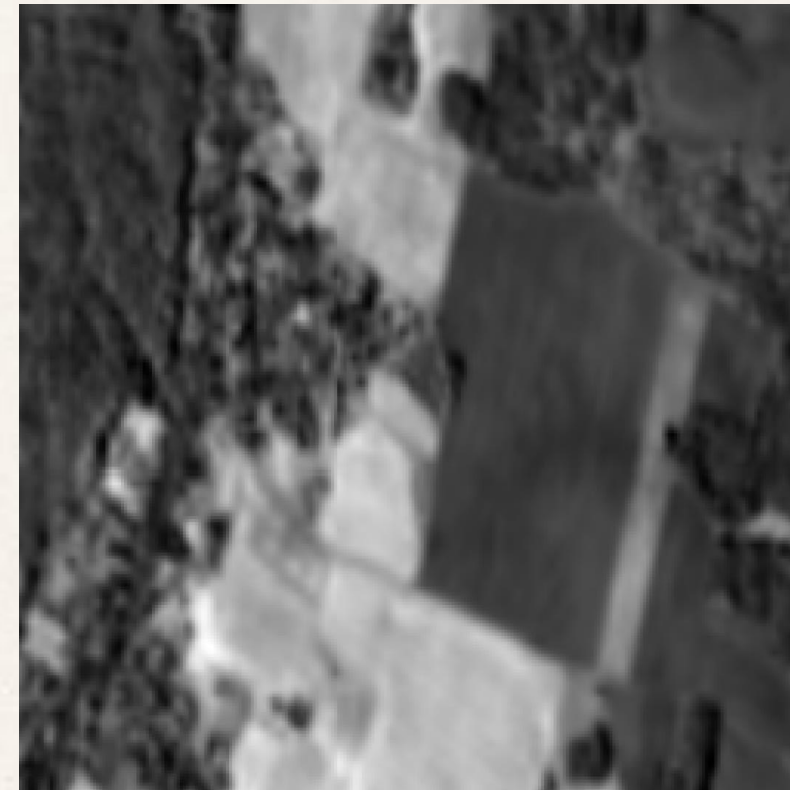


NIR



RGB

Window B



NIR



RGB

(Window a and b refer to different growing seasons which are combined when fed to the model.)

Data

Baseline Dataset that We Work With

Dataset Masks

Bolivia



Germany



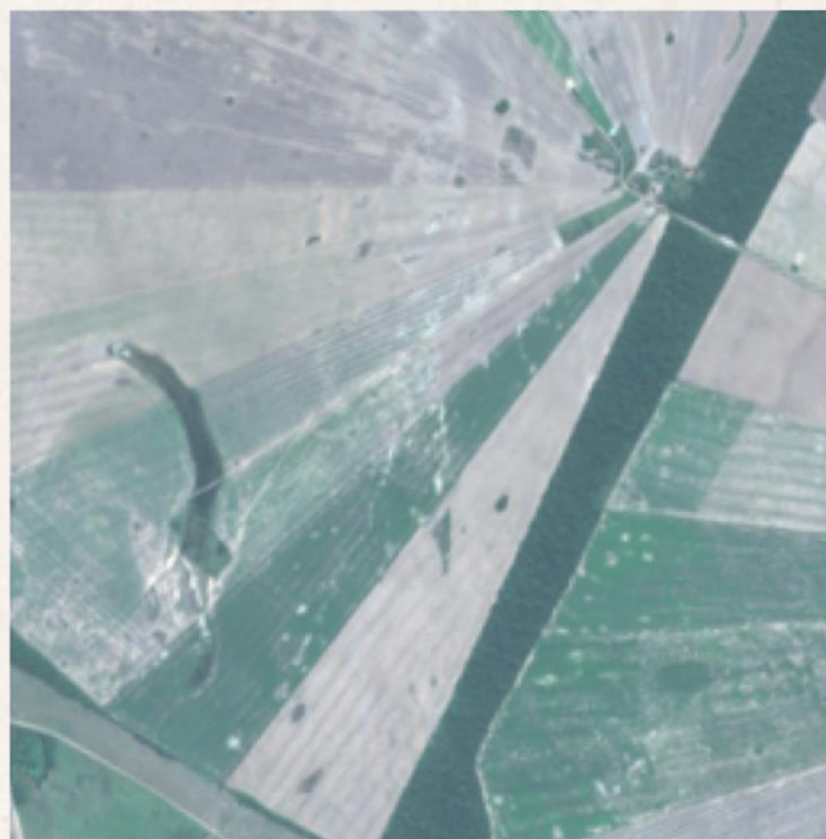


Geometric and “Strange” Data Augmentations

Geometric: Rotate, Flip and Transpose



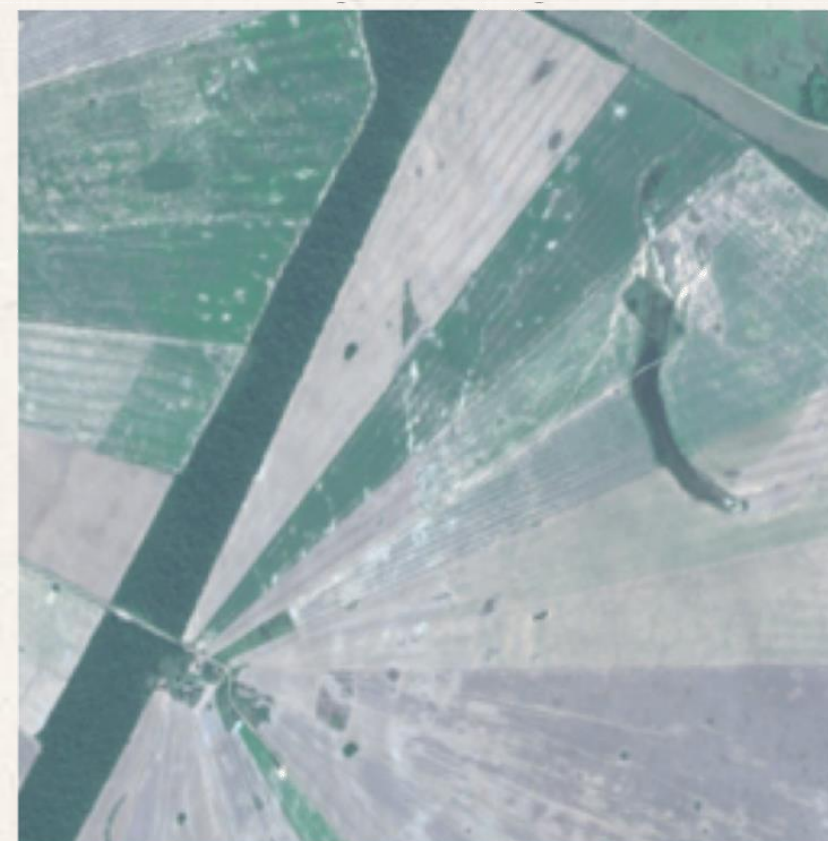
Ground Truth Image



Label Mask



Augmented Image



Label Mask

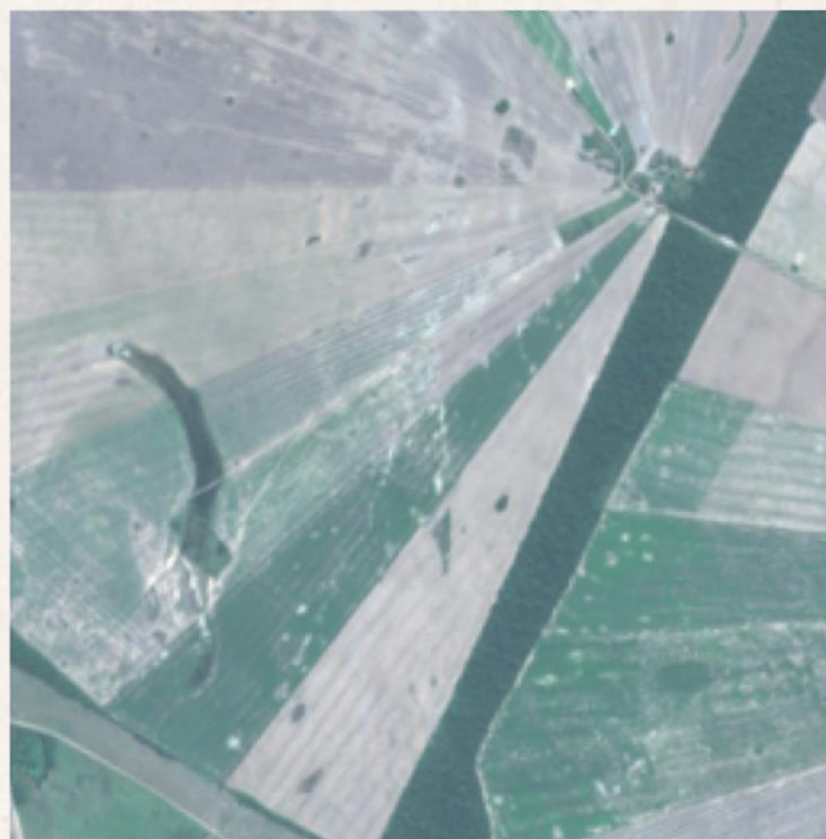


Geometric and “Strange” Data Augmentations

Strange: Dropout and Distortion



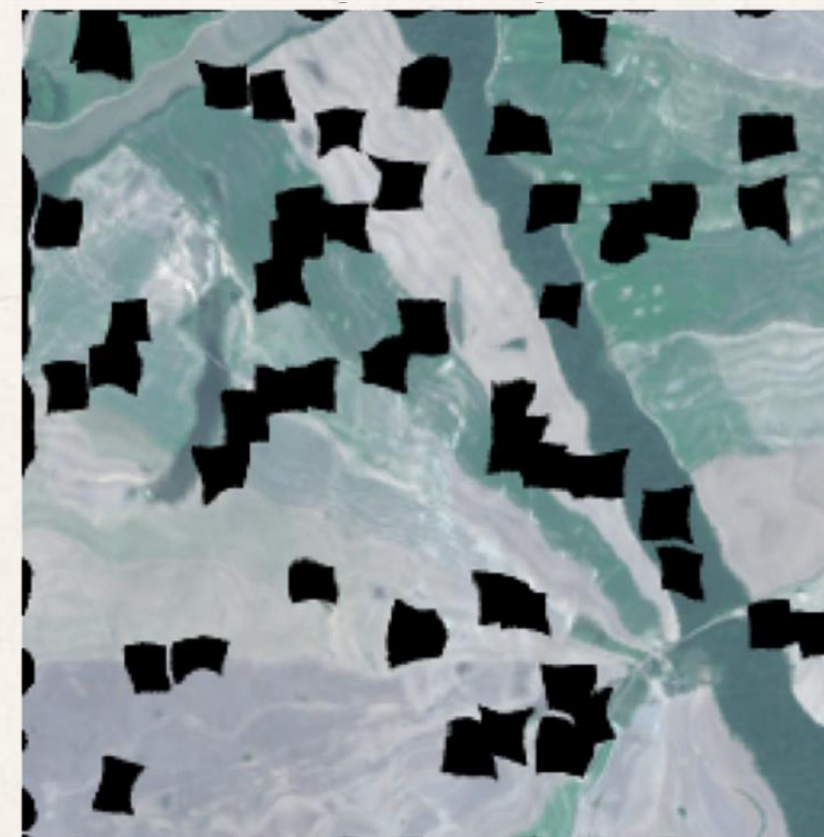
Ground Truth Image



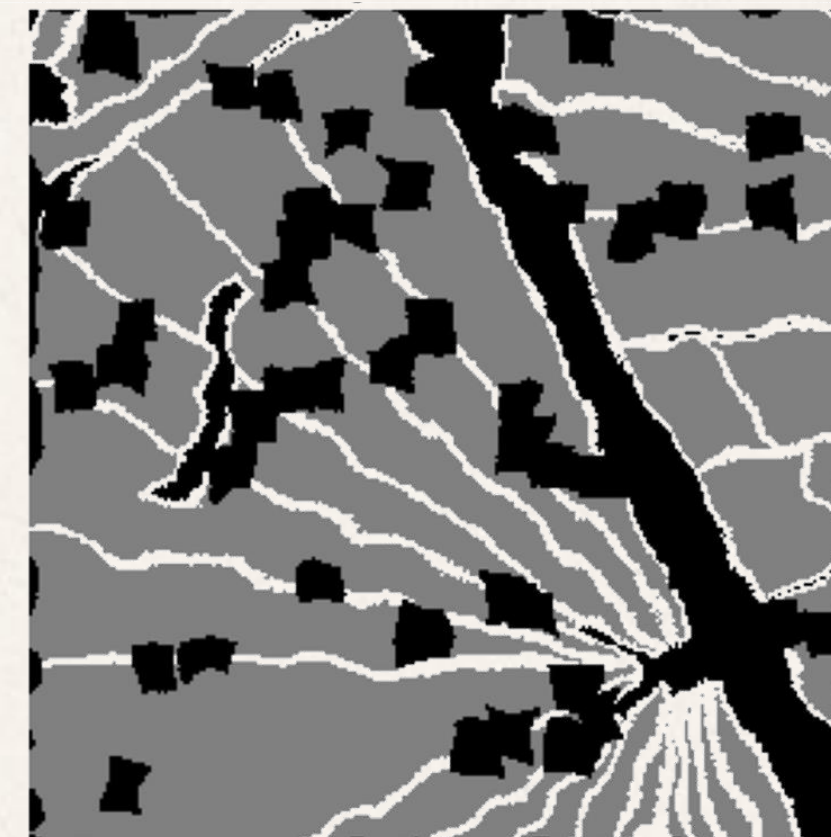
Label Mask



Augmented Image



Label Mask

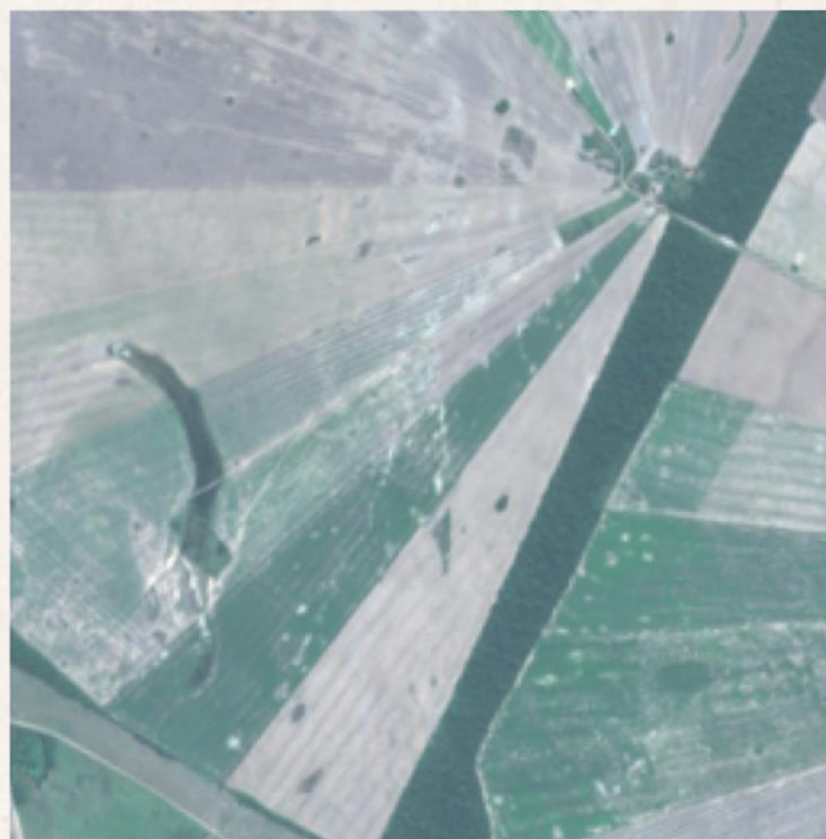


Geometric and “Strange” Data Augmentations

Combined Transformations



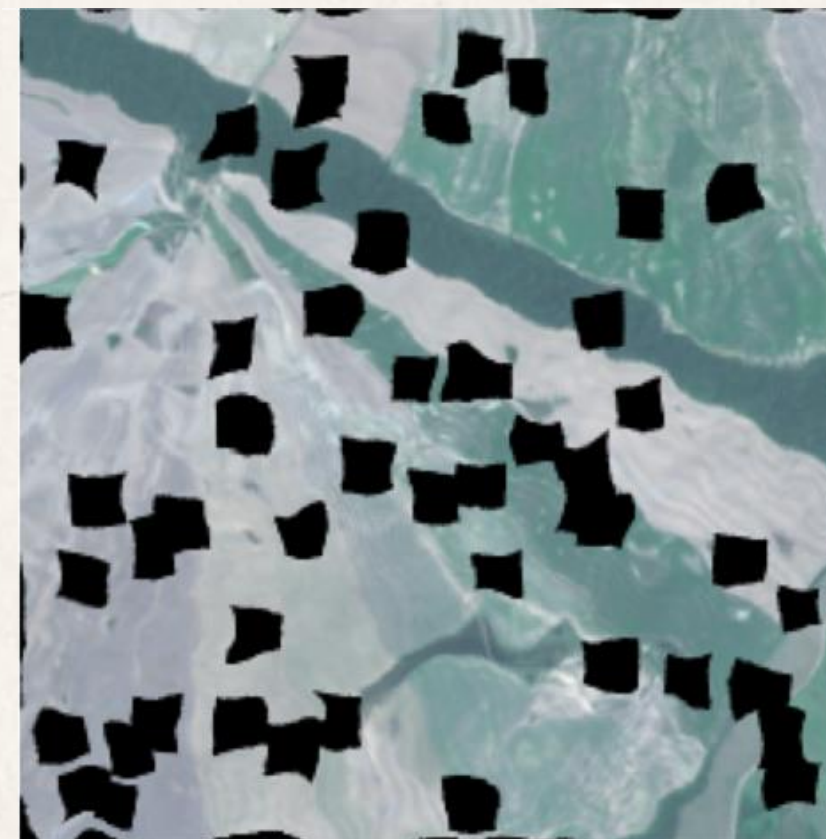
Ground Truth Image



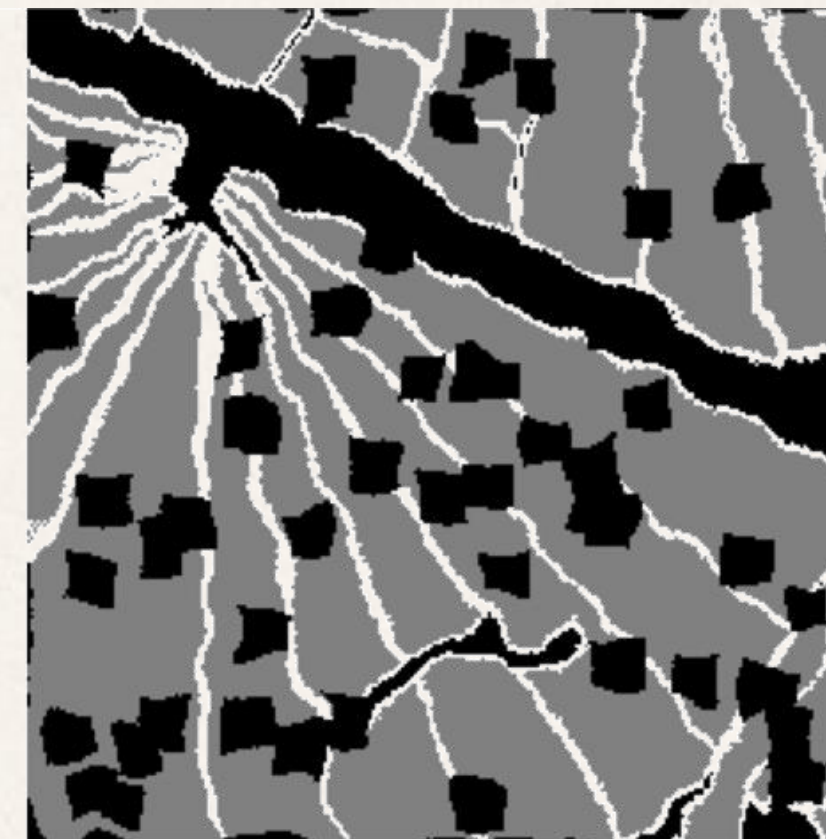
Label Mask



Augmented Image



Label Mask

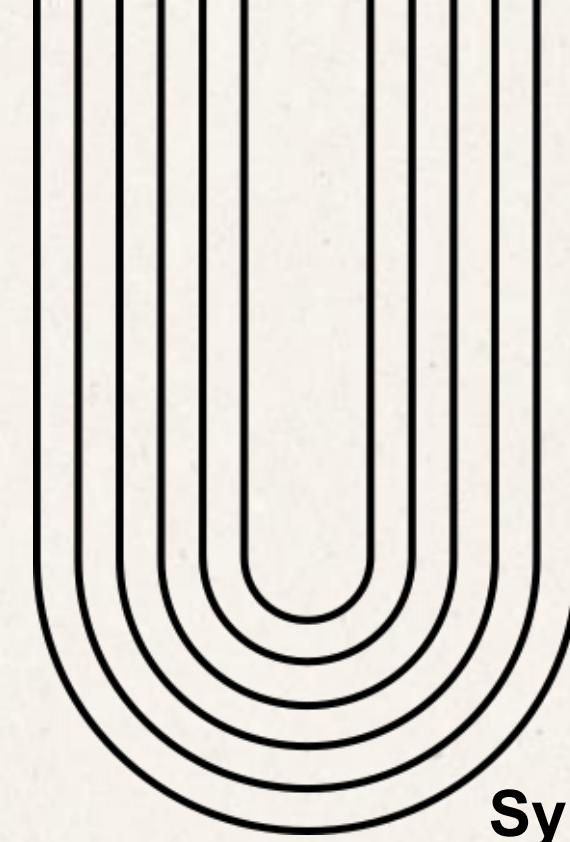


Generative Data Augmentations

Using SynthSeg's Generative Features

Ground Truth Image

Label Mask



SynthSeg's Image Generator

Extract **a series of values from the Ground Truth Image by Band and Label Mask**, including maximum value (95% percentile), minimum value (5% percentile), standard deviation to inform the image generator a broad range of values to select from.

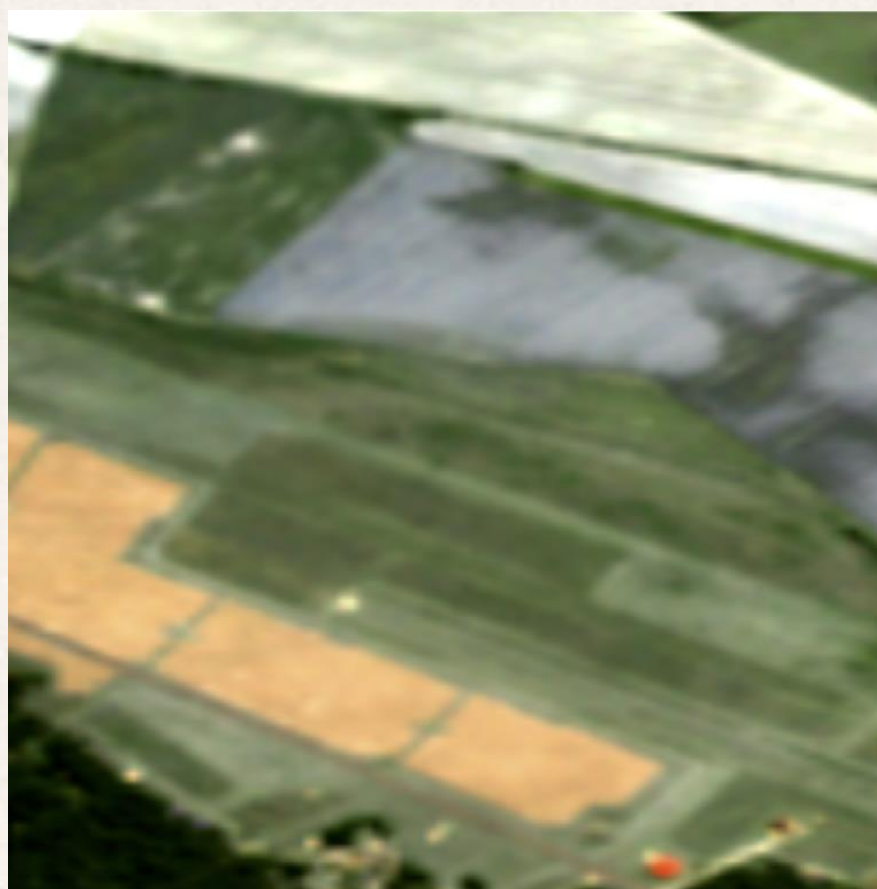


- Random flipping (disabled)
- Random affine (disabled)
- Random elastic (disabled)
- Random cropping (disabled)
- Random jittering (disabled)
- GMM Sampling (use given value ranges to determine generated pixel values)
- Add bias/noise
- Add Blur (disabled)

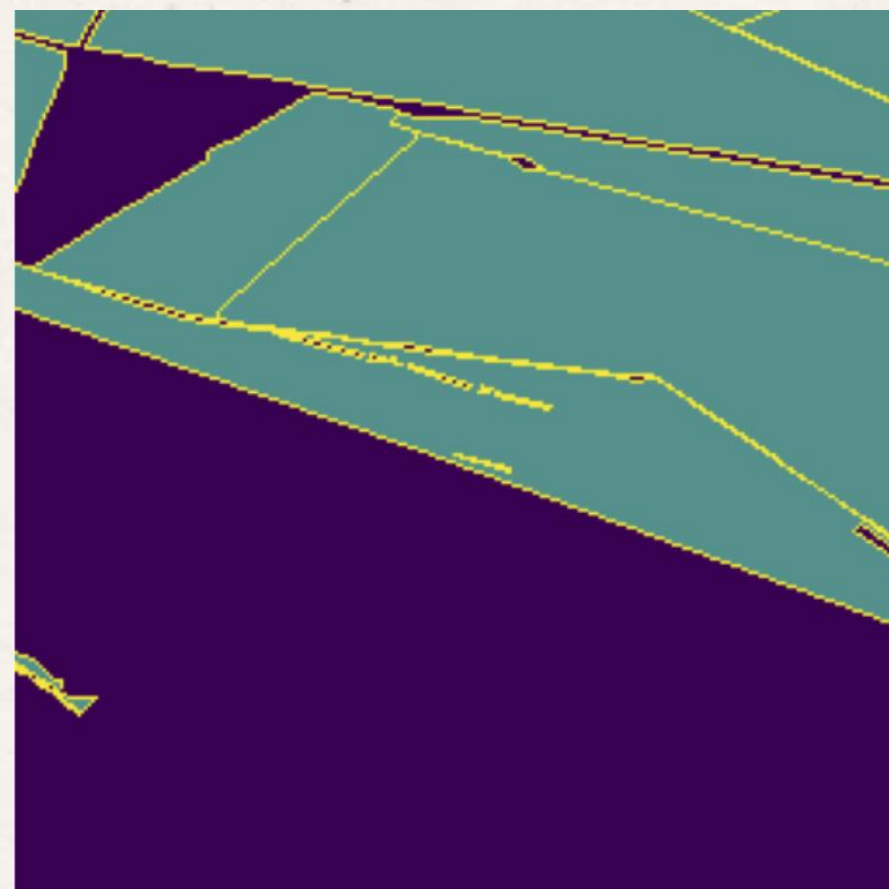
Generative Data Augmentations

Using SynthSeg's Generative Features

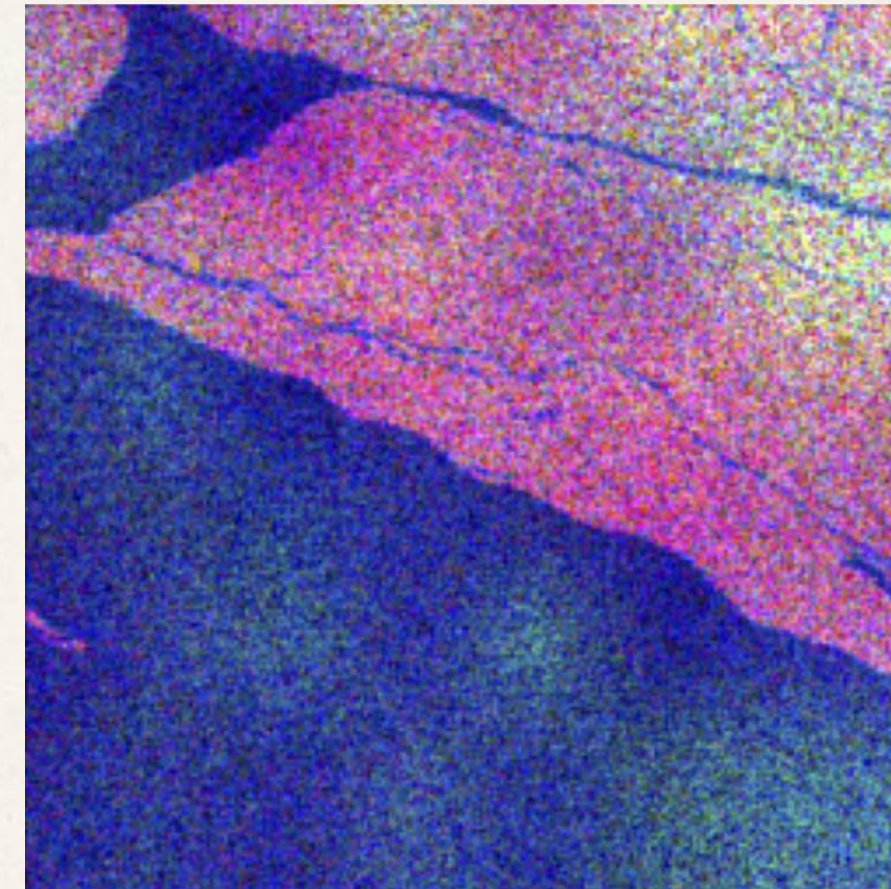
Ground Truth Image



Label Mask



Generated Image



Generative Data Augmentations

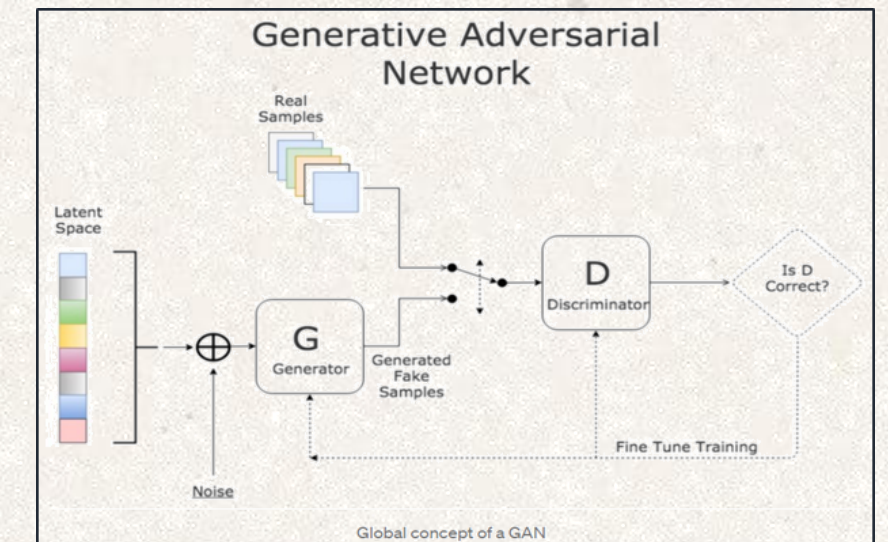
Using Pix2Pix Generative Adversarial Networks

Ground Truth Image

Label Mask

Pix2Pix GAN

In a Pix2Pix Generative Adversarial Networks (GAN) Model, Ground Truth Images are paired with Label Masks to be studied in a deep learning network.

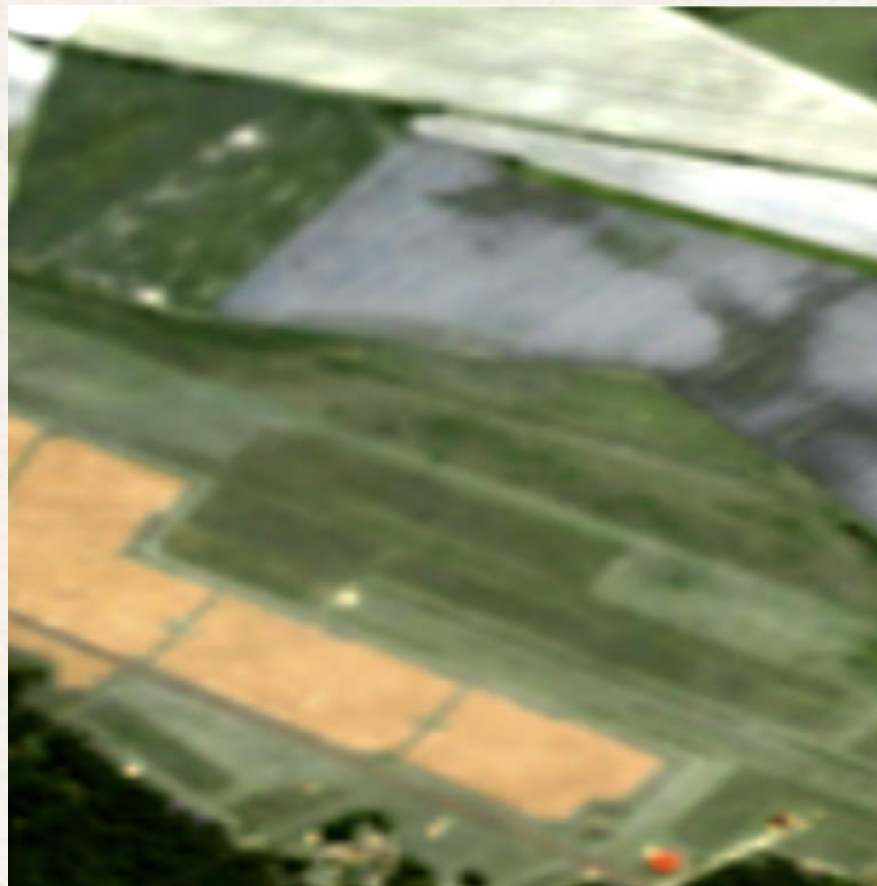


- U-Net-style generator + Discriminator

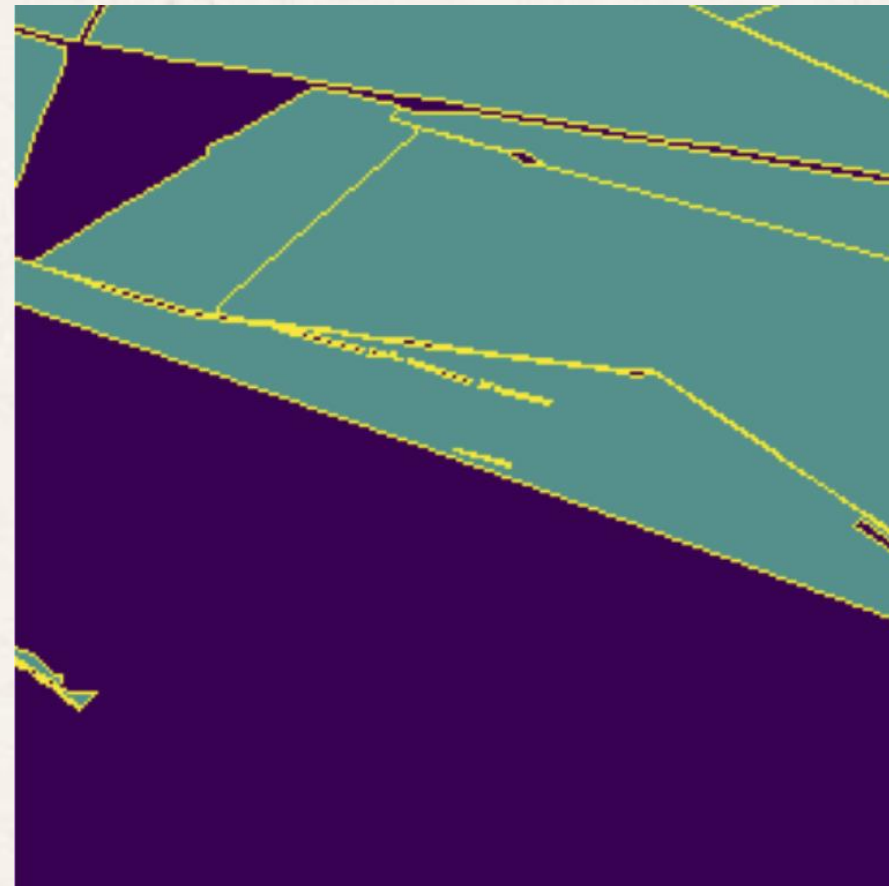
Generative Data Augmentations

Using Pix2Pix Generative Adversarial Networks

Ground Truth Image



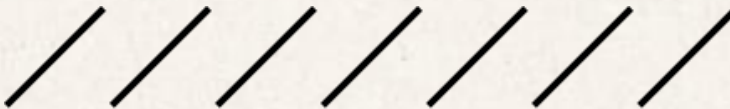
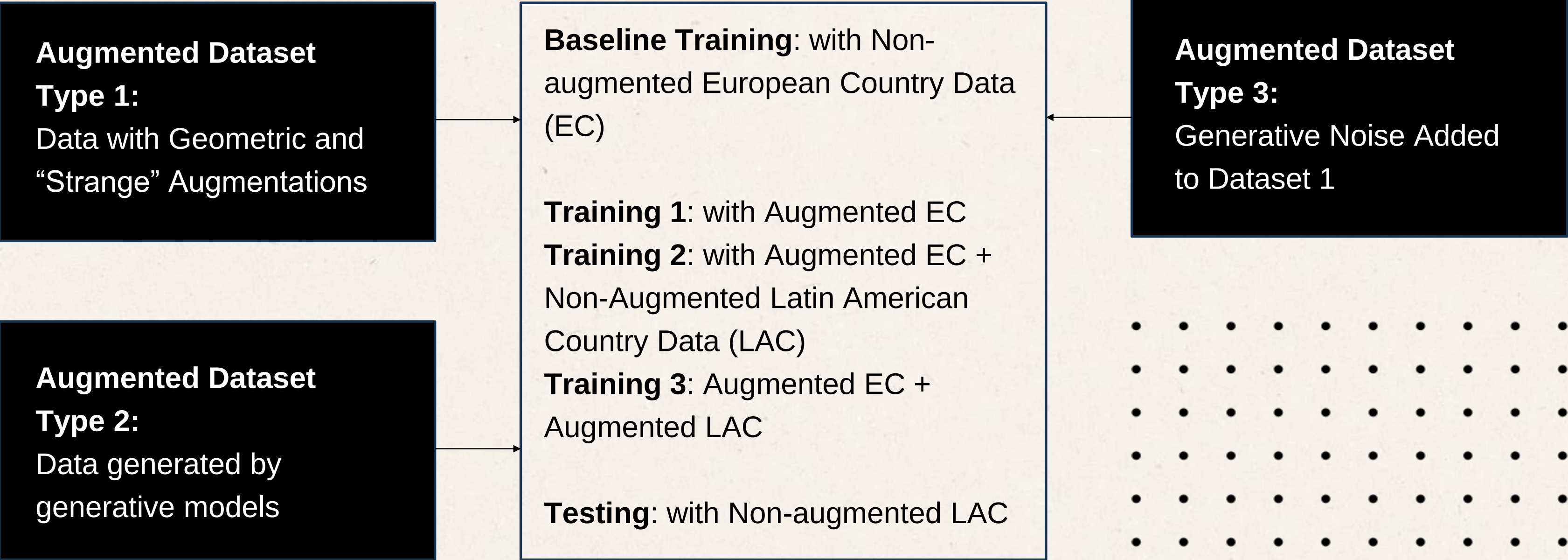
Label Mask



Generated Image

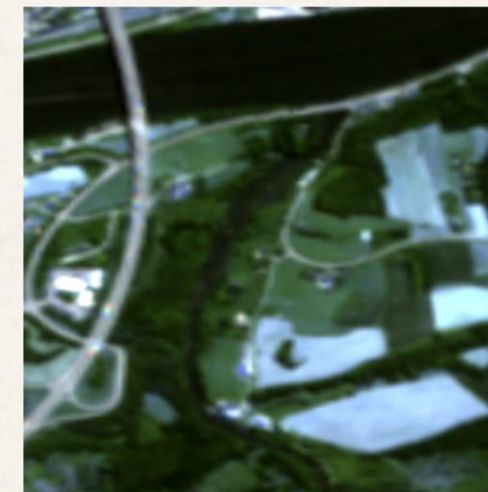
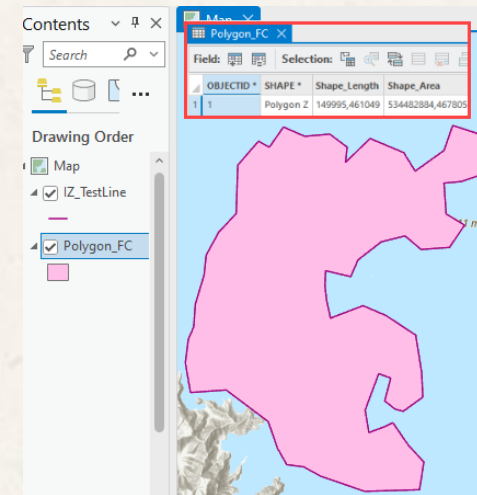
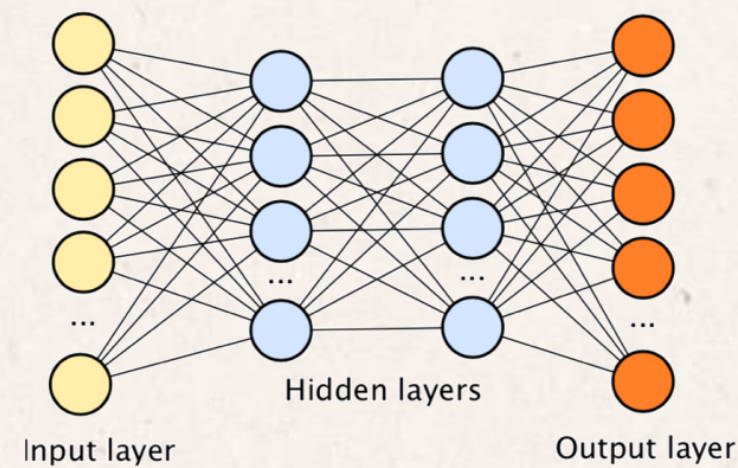


Dataset Analysis: 4 Experiments



Future Steps

Potential things we could do to further enhance model performance



Carry Out Experiments

Finish Planned Experiments to Assess Augmented Dataset Quality



Adding More Hand-Labeled Data

Adding More Ground Truth Data to Enhance the Quality of Synthetic Data



Introducing Data From More Countries

Adding More Geometric and Atmospheric Diversity to Dataset